

Experiment 1: PID

Code :

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

Output:

```

polisettysreeharsha@SREEHA  X  +  v
polisettysreeharsha@SREEHARSHA:~$ nano practical1.c
polisettysreeharsha@SREEHARSHA:~$ gcc practical1.c -o practical1
polisettysreeharsha@SREEHARSHA:~$ ./practical1
Enter linux command: lscpu
Parent PID : 598
Child PID : 599
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:         0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                  6
Model:                       186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4992.00
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):

```

```

polisettysreeharsha@SREEHA x + v
Caches (sum of all):
  L1d: 288 KiB (6 instances)
  L1i: 192 KiB (6 instances)
  L2: 7.5 MiB (6 instances)
  L3: 12 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
Child process completed.
polisettysreeharsha@SREEHARSHA:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
polisettysreeharsha@SREEHARSHA:~$ ps

```

```

Vmscape: Not affected
Child process completed.
polisettysreeharsha@SREEHARSHA:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
polisettysreeharsha@SREEHARSHA:~$ ps
  PID TTY          TIME CMD
  333 pts/0        00:00:00 bash
  619 pts/0        00:00:00 ps
polisettysreeharsha@SREEHARSHA:~$ |

```

Experiment 2: COPYING CONTENT OF ONE FILE FROM ANOTHER

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

Output:

```
polisettysreeharsha@SREEHARSHA:~$ ./practical2
File copied successfully.
polisettysreeharsha@SREEHARSHA:~$ cat output.txt
This is a test file.
polisettysreeharsha@SREEHARSHA:~$ ./copy
File copied successfully.
polisettysreeharsha@SREEHARSHA:~$ |
```